

Chi-Wei HU

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Aspiring software engineer graduated from MS in Computer Science at USC. Proficient in C#, C++ and Python, with project experience in object-oriented programming, database management, and AI/ML pipeline application. Actively seeking a software engineer position.

Education

University of Southern California(USC)

MS in Computer Science

Los Angeles, U.S.

Sept. 2024 - May 2026

- Coursework: Multimedia System Design (CSCI576), Foundation of Artificial Intelligence (CSCI561), Machine Learning for Data Science (CSCI552), Operating System (CSCI402), Analysis of Algorithm (CSCI570), Database Systems (CSCI585)

National Taiwan University (NTU)

MS in Biochemistry

Taipei, Taiwan

Sept. 2020 - July 2022

- Thesis: Engineering of A Multi-cell Type Immunotherapy leveraging synNotch-driven Macrophages
- GPA: 3.88/4.3

National Cheng Kung University (NCKU)

BA in Life Science

Tainan, Taiwan

Sept. 2017-July 2020

Skills

- **Languages:** Python, C#, C++, C, Rust, SQL
- **Backend & Infra:** Docker, RPC & IDL-based service interfaces (Apache Thrift), Flask, Express.js, MySQL, MongoDB, Bash/Linux, CI/CD, multithreaded programming & synchronization
- **ML/AI:** scikit-learn, NumPy/Pandas, CLIP, Whisper, TransNetV2, Ollama (local LLM), Claude API, multimodal fusion, model evaluation (subject-level LOSO, ROC-AUC)
- **Signal & Media Processing:** Digital Signal Processing (DCT/IDCT), Image Processing (JPEG compression)
- **Version Control & Code Review:** Git, Mercurial, Phabricator/Diff
- **Embedded & Industrial Automation:** WPF, firmware, RS-232/RS-485, SEMI SECS/GEM, kernel programming (C)

Experience

Software Engineer Intern, WhatsApp: Small Business & CoExistence

Meta Inc.

Menlo Park, California

May 2025- Aug. 2025

- Built a Rust-based CLI debugging tool on top of the Apache Thrift framework, enabling cross-language backend APIs to analyze records from ZippyDB and improving issue triaging
- Establish webhook replay functionality in message echo workflow of WhatsApp Business CloudAPI platform to ensure reliable cross-system message delivery

Automation Engineer, Department of Computer-Integrated Manufacturing (CIM)

Strong Electro-Mechanical Co., Ltd

Tainan, Taiwan

Sept. 2022- May 2024

- Developed a Standalone Automation Process for etching machines in C#, utilizing SECS for device communication and improving process efficiency by 20%
- Established a C++ monitoring system for exposure machines, enabling communication with embedded systems like Arduino and DAQ devices and integrating with MySQL, resulting in a 15% reduction in downtime

Projects & Research Experience

ParkScreen — Multimodal Parkinson's Screening Tool [\(Github\)](#) [\(Link\)](#)

Python, scikit-learn, Claude API, Docker

- Built with Claude: Life Sciences Hackathon, 2026 (Anthropic x Gladstone x Cerebral Valley)
- A clinical decision-support tool that screens for Parkinson's disease from three task-matched user recordings
- Fused per-channel logistic regressions on vocal and facial-AU signals via score-level late fusion (AUC 0.758, subject-level LOSO on 95 NeuroVoz subjects), then piped the fused scores into Claude Opus 4 to generate structured clinical-style reports

Multimodal Video Content Segmentation System [\(Github\)](#)

Python, CLIP, Whisper, Ollama, TransNetV2

- Architected a multi-modal pipeline that temporally segments videos into content vs. non-content regions by fusing visual, speech, and LLM signals; scored full marks on all final test cases
- Owned the fusion engine: weighted voting across modalities, dual-path (speech vs. non-speech) classification, and context-aware confidence calibration